Compressor off-design report: summary and diagnostics

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Rotor-Rotor Same Shaft  |  run: Max_PR_Compressor_design_rr_same_shaft

Rotor-Rotor Same Shaft  |  PR_des = 2.0095, eta_des = 0.9170  |  correlations: polynomial, Howell factor 2.00

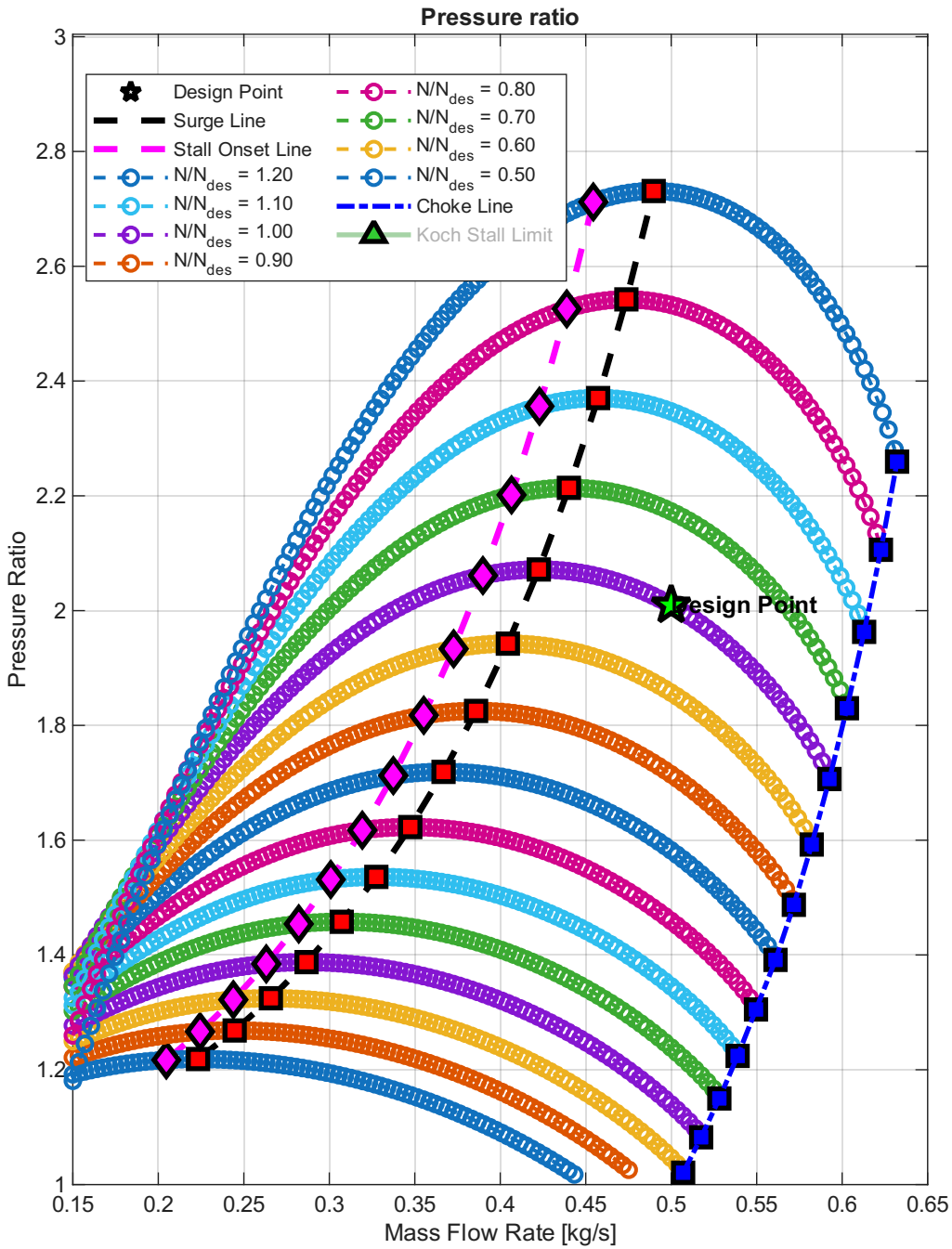
STABILITY (design speed): first limit = peak at m = 0.4226 kg/s (PR 2.072); stall margin SM = 22.0 %
   peak of the speed line (zero slope): m = 0.4226 kg/s | Howell stall onset: m = 0.3899 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5926 kg/s (Rotor 1 exit (relative)) -> choke margin 18.5 %, operating range (m_choke - m_lim)/m_choke = 28.7 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.413 -> surge if the installation has V_p/(A_c L_c) > 3.75 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Rotor 1 (omega/omega_des = 1.53), lowest de Haller Rotor 1 (0.53). Details: Tables > Blade
      rows.

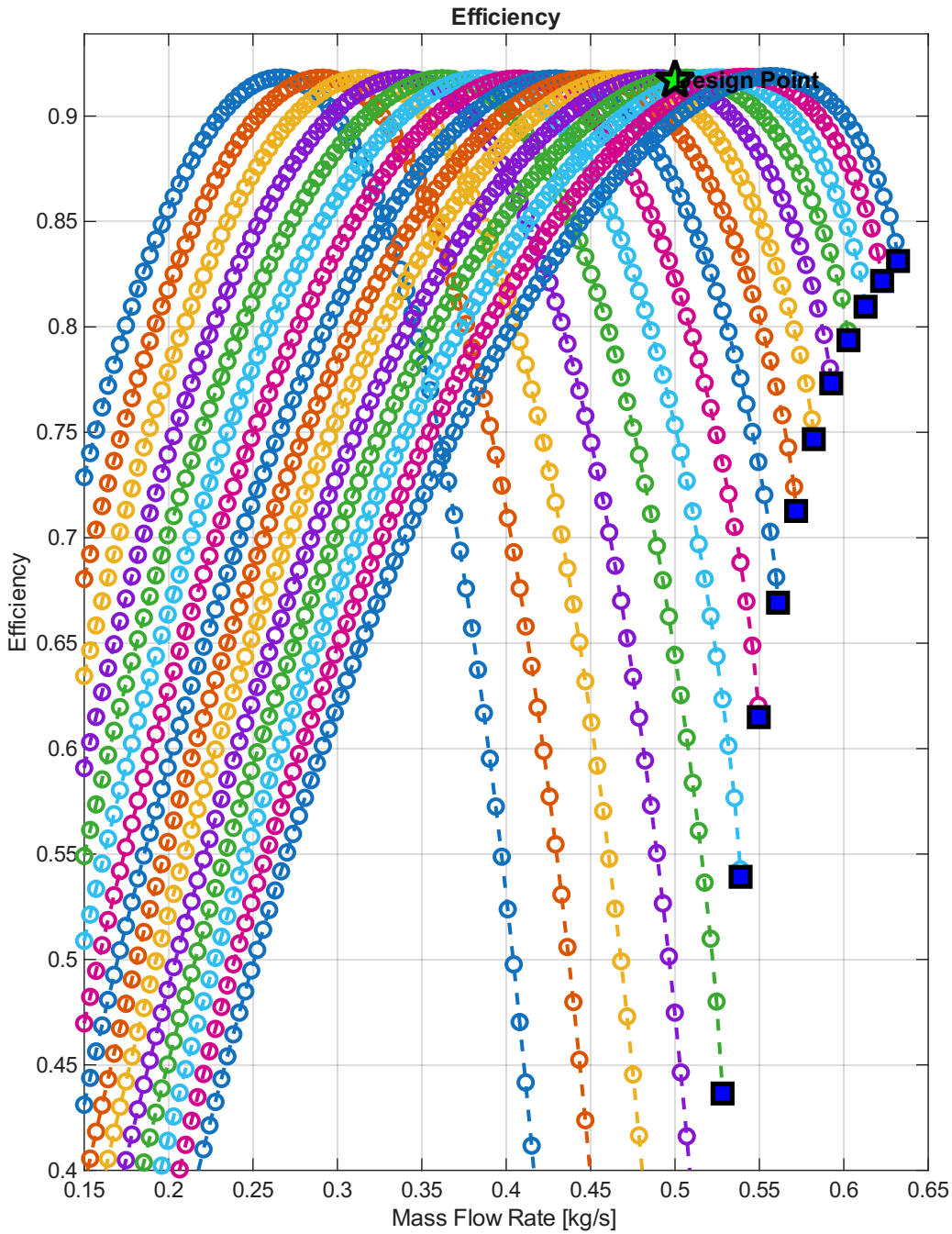
KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 4 of 15 speed lines (N/N_des 1.05 to 1.20).
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in the same rotating frame (same shaft), so they
      are not re-energized; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: LOW MARGIN (0.89, target 0.85)
     - RAISE the solidity x1.15 (stage L/g2 2.13 -> 2.45): R1 1.97 -> 2.26, R2 1.42 -> 1.63; re-run the design script with it (profile loss changes)
      [solidity above 2: outside the Carter/NACA-65 range of the design scripts]  [L/g2 outside Koch's data 0.6-2.2]
     - or LOWER the stage static-pressure rise by about 5% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.82
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.84
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
      1 the ratio would be 0.70 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: F_ef = [1.08 0.25] (data 0.39-1.11) -> indicative only
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Inputs of this run (.mat data and settings)

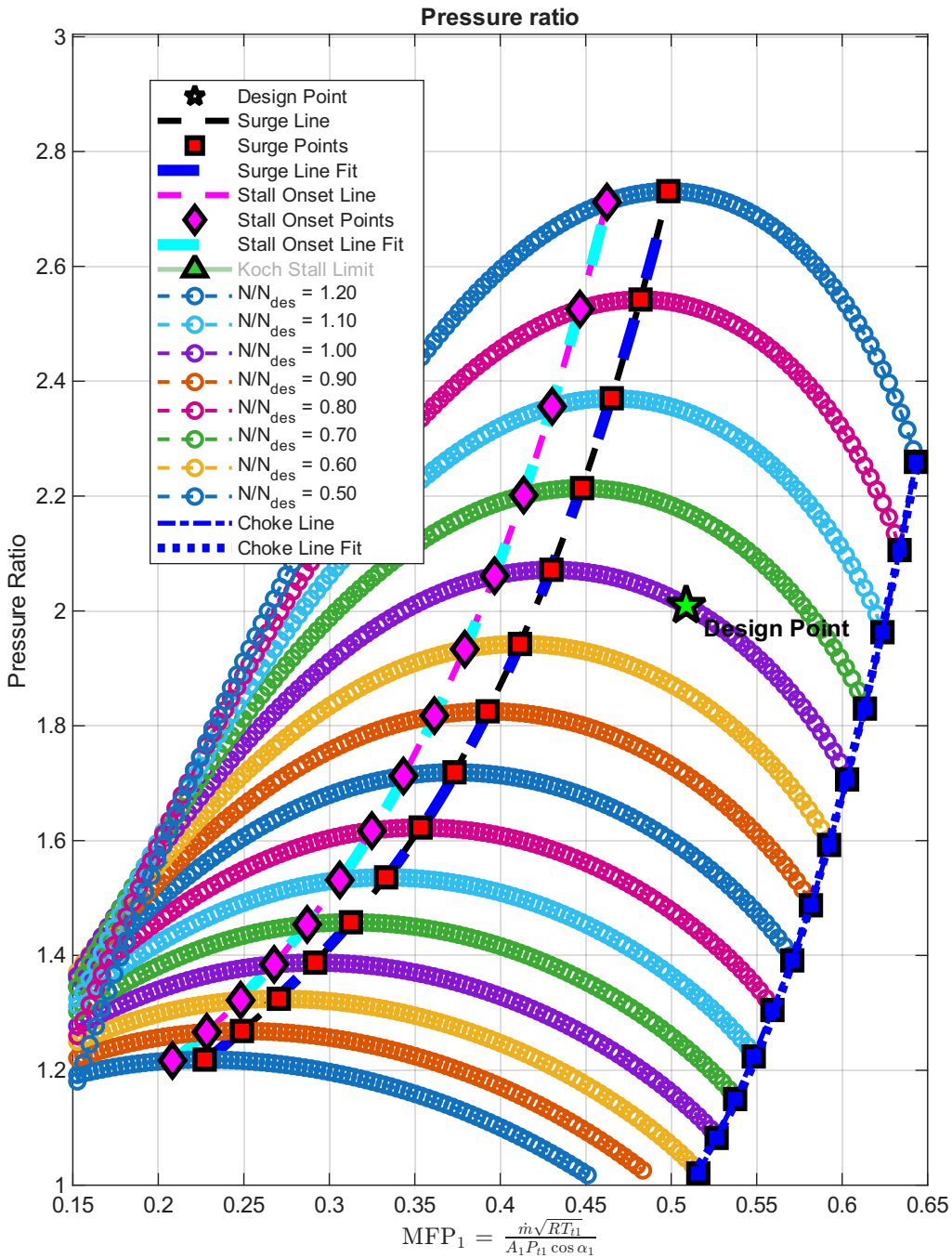
Variable	Value	Unit	Source
Configuration	Rotor-Rotor same shaft		
Input source	.mat: Max_PR_Compressor_design_rr_s...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rr_s...
n_stages	2	-	.mat: Max_PR_Compressor_design_rr_s...
A_vec	0.00360564 0.00323629 0.00313338	m^2	.mat: Max_PR_Compressor_design_rr_s...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rr_s...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rr_s...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rr_s...
w_loss_rotor	0.141203	-	.mat: Max_PR_Compressor_design_rr_s...
beta3_dash	38.2935	deg	.mat: Max_PR_Compressor_design_rr_s...
beta4_dash	-12.5679	deg	.mat: Max_PR_Compressor_design_rr_s...
deviation_1	3.53466	deg	.mat: Max_PR_Compressor_design_rr_s...
deviation_2	9.07861	deg	.mat: Max_PR_Compressor_design_rr_s...
incidence_1	6.87796	deg	.mat: Max_PR_Compressor_design_rr_s...
incidence_2	0.109785	deg	.mat: Max_PR_Compressor_design_rr_s...
deflection_1	20.3219	deg	.mat: Max_PR_Compressor_design_rr_s...
deflection_2	41.8926	deg	.mat: Max_PR_Compressor_design_rr_s...
w_loss_rotor2	0.0800466	-	.mat: Max_PR_Compressor_design_rr_s...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rr_s...
sigma_1	1.97032	-	.mat: Max_PR_Compressor_design_rr_s...
sigma_2	1.42267	-	.mat: Max_PR_Compressor_design_rr_s...
chord	0.01	m	.mat: Max_PR_Compressor_design_rr_s...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	141	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

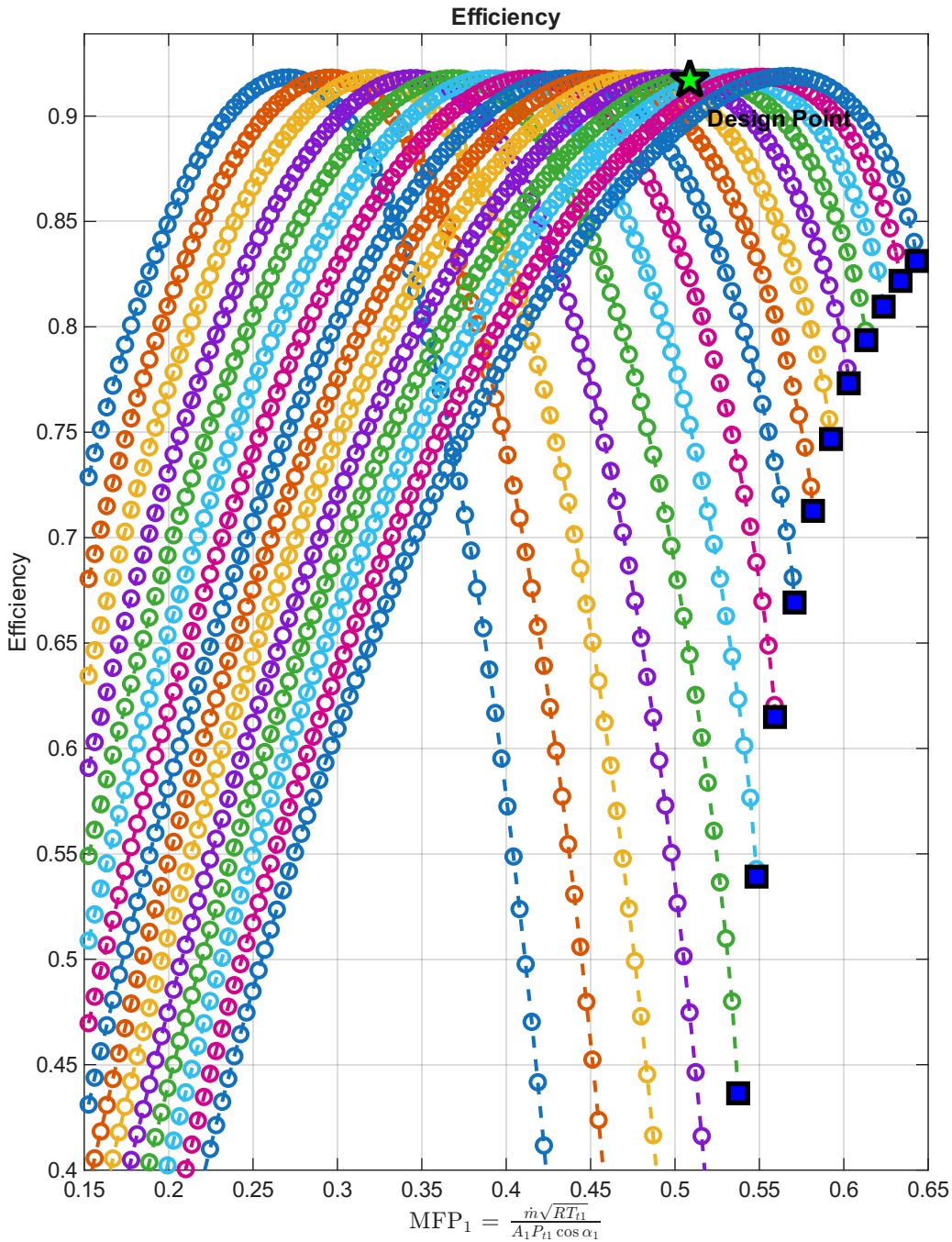
Standard (dimensional) map - Rotor-Rotor Same Shaft



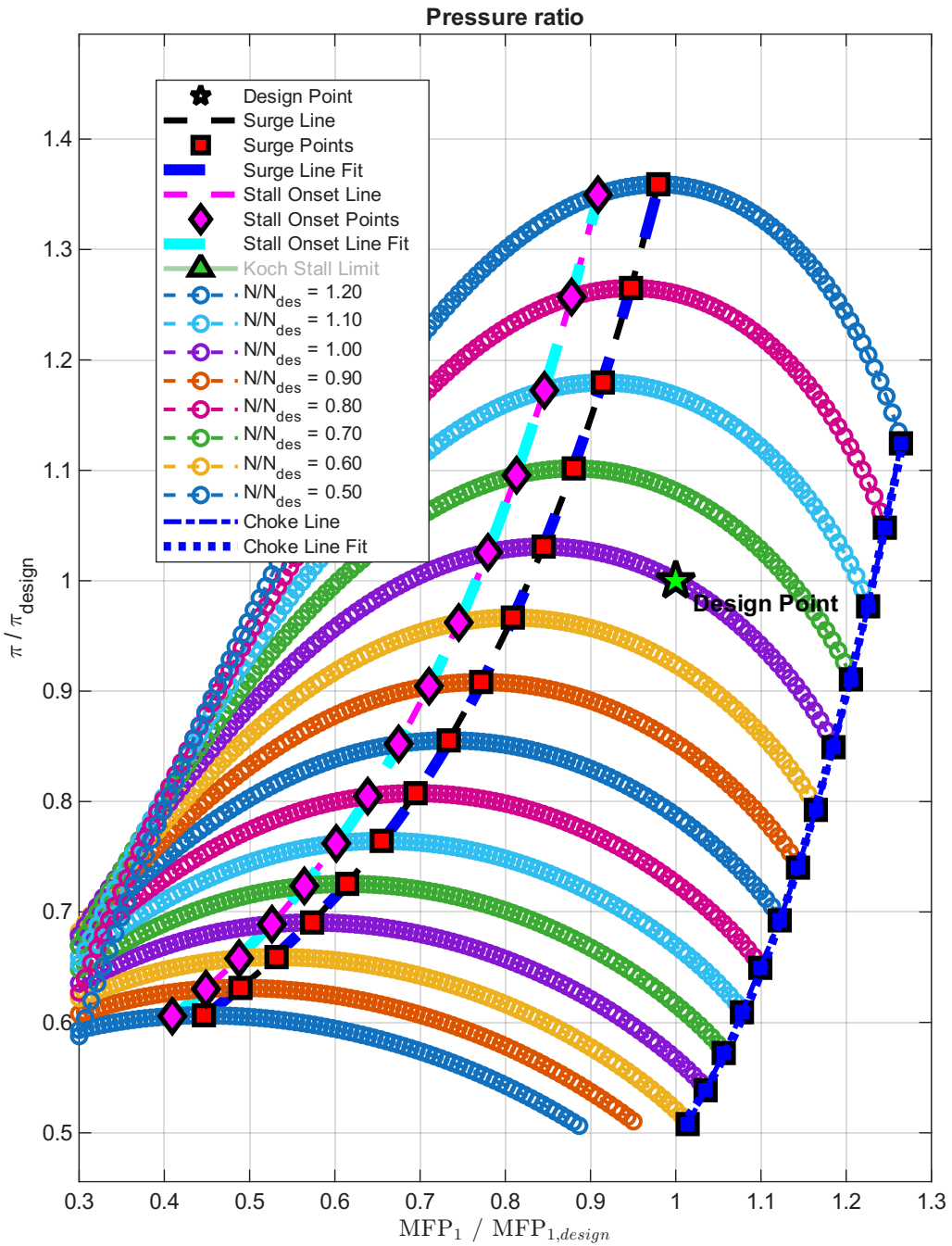


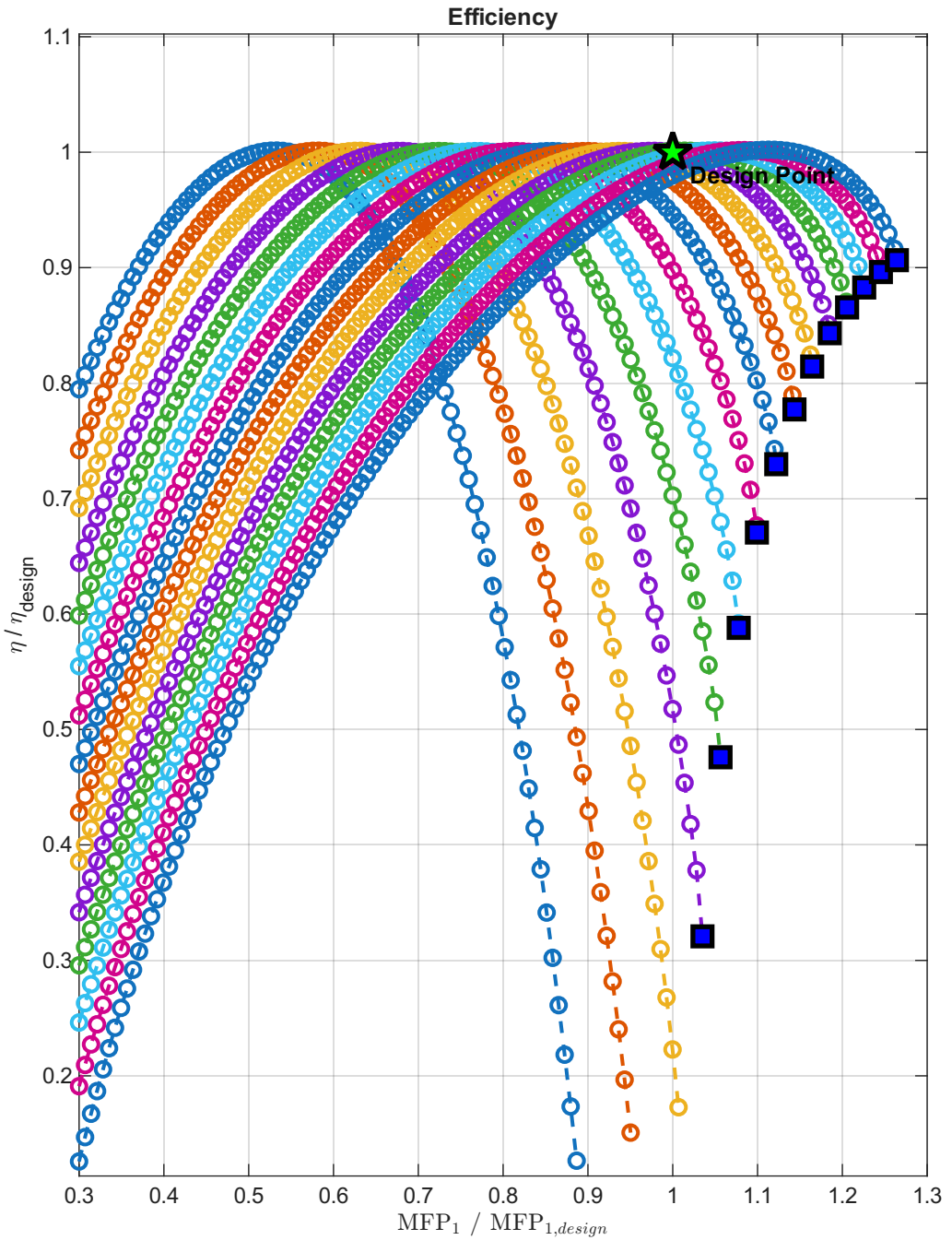
Non-dimensional map - Rotor-Rotor Same Shaft



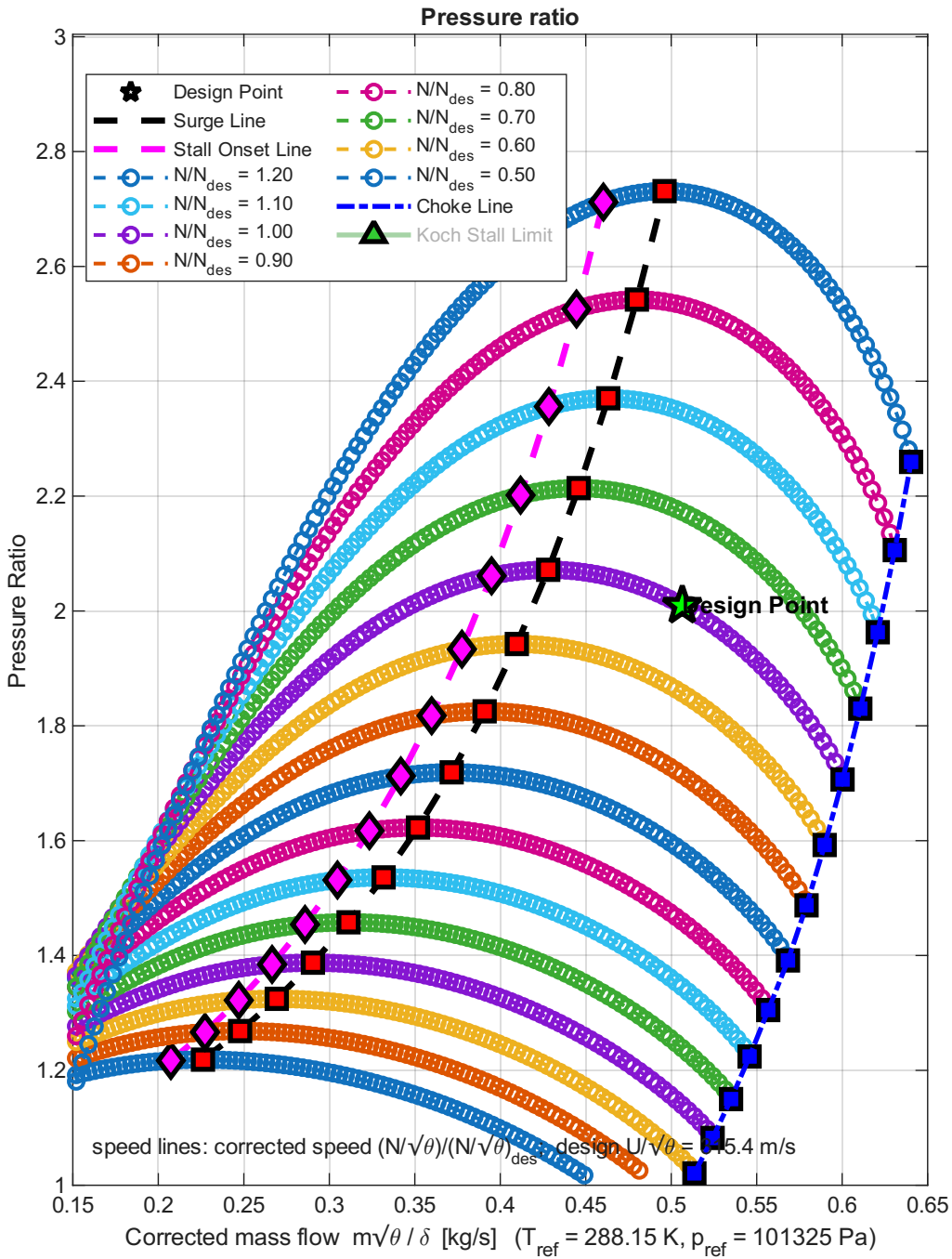


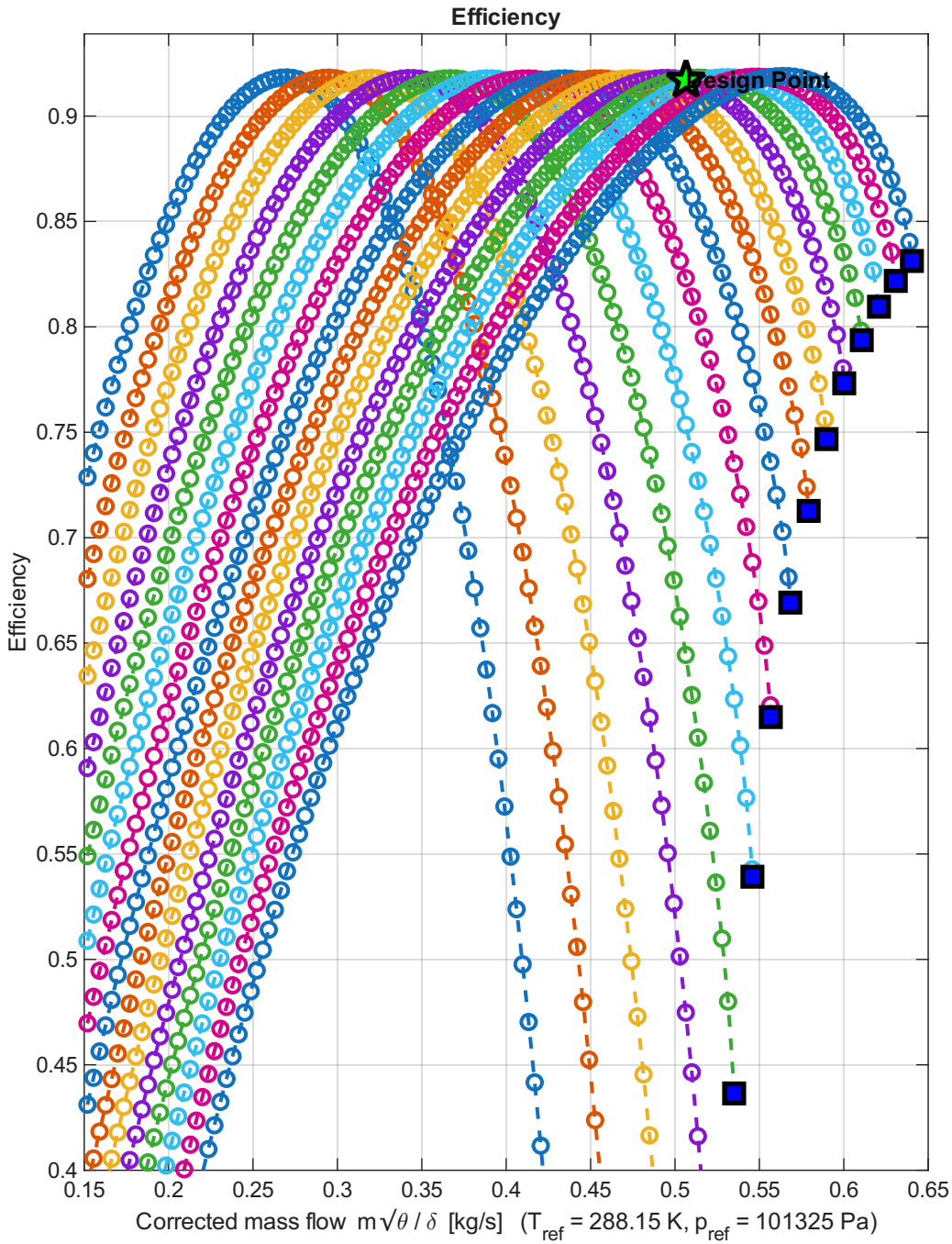
Relative non-dimensional map - Rotor-Rotor Same Shaft





Corrected map - Rotor-Rotor Same Shaft





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Same Shaft ====
Surge line:
Surge_PR = 41.8295*MFP^3 + -28.8275*MFP^2 + 9.2038*MFP + 0.1207   [R^2 = 0.99994]
Stall-onset line (Howell):
StallOnset_PR = 44.1361*MFP^3 + -26.2080*MFP^2 + 7.8540*MFP + 0.3158   [R^2 = 0.99997]
Choke line:
Choke_PR = 42.1410*MFP^2 + -39.3306*MFP + 10.1088   [R^2 = 0.99952]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Rotor Same Shaft ====
Surge line:
(PR/PR_design) = 2.7409*x^3 + -3.7129*x^2 + 2.3301*x + 0.0600   (x = MFP/MFP_design)   [R^2 = 0.99994]
Stall-onset line (Howell):
(PR/PR_design) = 2.8920*x^3 + -3.3756*x^2 + 1.9884*x + 0.1572   (x = MFP/MFP_design)   [R^2 = 0.99997]
Choke line:
(PR/PR_design) = 5.4277*x^2 + -9.9573*x + 5.0305   (x = MFP/MFP_design)   [R^2 = 0.99952]
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Design point

Quantity	Value
Configuration	Rotor-Rotor Same Shaft
Pressure ratio PR_des	2.0095
Isentropic efficiency eta_des	0.91701
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.50874
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.48649	0.495	0.914		Rotor 1 exit (relative)	no
0.55	0.49661	0.5053	0.965		Rotor 1 exit (relative)	no
0.6	0.507	0.5159	1.021	0.092	Rotor 1 exit (relative)	yes
0.65	0.51754	0.5266	1.082	0.294	Rotor 1 exit (relative)	yes
0.7	0.52817	0.5374	1.15	0.436	Rotor 1 exit (relative)	yes
0.75	0.53887	0.5483	1.224	0.539	Rotor 1 exit (relative)	yes
0.8	0.54962	0.5592	1.305	0.615	Rotor 1 exit (relative)	yes
0.85	0.56103	0.5708	1.391	0.669	Rotor 1 exit (relative)	yes
0.9	0.57176	0.5818	1.487	0.713	Rotor 1 exit (relative)	yes
0.95	0.58222	0.5924	1.592	0.747	Rotor 1 exit (relative)	yes
1	0.59256	0.6029	1.707	0.773	Rotor 1 exit (relative)	yes
1.05	0.6028	0.6133	1.83	0.794	Rotor 1 exit (relative)	yes
1.1	0.61288	0.6236	1.963	0.81	Rotor 1 exit (relative)	yes
1.15	0.6227	0.6336	2.106	0.822	Rotor 1 exit (relative)	yes
1.2	0.63215	0.6432	2.26	0.831	Rotor 1 exit (relative)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.20467	1.217	0.865	Rotor 1
0.55	0.22438	1.266	0.864	Rotor 1
0.6	0.24387	1.322	0.864	Rotor 1
0.65	0.26313	1.384	0.864	Rotor 1
0.7	0.28214	1.454	0.863	Rotor 1
0.75	0.30087	1.532	0.863	Rotor 1
0.8	0.31932	1.617	0.863	Rotor 1
0.85	0.33746	1.713	0.862	Rotor 1
0.9	0.35527	1.818	0.862	Rotor 1
0.95	0.37275	1.933	0.862	Rotor 1
1	0.38987	2.061	0.861	Rotor 1
1.05	0.40661	2.202	0.861	Rotor 1
1.1	0.42296	2.356	0.861	Rotor 1
1.15	0.43891	2.526	0.861	Rotor 1
1.2	0.45444	2.712	0.861	Rotor 1

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.22322	1.219	0.892	-0.01854	peak (zero slope)
0.55	0.24465	1.268	0.892	-0.02027	peak (zero slope)
0.6	0.26583	1.325	0.892	-0.02196	peak (zero slope)
0.65	0.28672	1.388	0.892	-0.02359	peak (zero slope)
0.7	0.30729	1.458	0.892	-0.02515	peak (zero slope)
0.75	0.32752	1.536	0.891	-0.02664	peak (zero slope)
0.8	0.34737	1.623	0.891	-0.02806	peak (zero slope)
0.85	0.36683	1.719	0.891	-0.02938	peak (zero slope)
0.9	0.38587	1.825	0.891	-0.0306	peak (zero slope)
0.95	0.40446	1.942	0.891	-0.03171	peak (zero slope)
1	0.42257	2.072	0.891	-0.0327	peak (zero slope)
1.05	0.44017	2.214	0.891	-0.03357	peak (zero slope)
1.1	0.45726	2.37	0.891	-0.03429	peak (zero slope)
1.15	0.47379	2.542	0.891	-0.03488	peak (zero slope)
1.2	0.48976	2.731	0.892	-0.03532	peak (zero slope)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	NaN	NaN	NaN	not reached
0.55	NaN	NaN	NaN	not reached
0.6	NaN	NaN	NaN	not reached
0.65	NaN	NaN	NaN	not reached
0.7	NaN	NaN	NaN	not reached
0.75	NaN	NaN	NaN	not reached
0.8	NaN	NaN	NaN	not reached
0.85	NaN	NaN	NaN	not reached
0.9	NaN	NaN	NaN	not reached
0.95	NaN	NaN	NaN	not reached
1	NaN	NaN	NaN	not reached
1.05	0.47811	2.198	0.913	Rotors R1+R2
1.1	0.50904	2.336	0.917	Rotors R1+R2
1.15	0.53607	2.484	0.918	Rotors R1+R2
1.2	0.56082	2.643	0.919	Rotors R1+R2

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	peak	0.22322	1.219	157.7	351.1	0.2246	12.69	NaN	-
0.55	peak	0.24465	1.268	173.4	353.3	0.2454	10.62	NaN	-
0.6	peak	0.26583	1.325	189.2	355.8	0.2659	9.05	NaN	-
0.65	peak	0.28672	1.388	205	358.4	0.286	7.83	NaN	-
0.7	peak	0.30729	1.458	220.7	361.2	0.3056	6.85	NaN	-
0.75	peak	0.32752	1.536	236.5	364.2	0.3247	6.07	NaN	-
0.8	peak	0.34737	1.623	252.3	367.4	0.3433	5.43	NaN	-
0.85	peak	0.36683	1.719	268.1	370.7	0.3615	4.9	NaN	-
0.9	peak	0.38587	1.825	283.8	374.3	0.3792	4.45	NaN	-
0.95	peak	0.40446	1.942	299.6	377.9	0.3963	4.07	NaN	-
1	peak	0.42257	2.072	315.4	381.8	0.413	3.75	NaN	-
1.05	peak	0.44017	2.214	331.1	385.8	0.4292	3.47	NaN	-
1.1	peak	0.45726	2.37	346.9	389.9	0.4448	3.23	NaN	-
1.15	peak	0.47379	2.542	362.7	394.2	0.46	3.02	NaN	-
1.2	peak	0.48976	2.731	378.4	398.6	0.4747	2.84	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	6.88	0	0.1412	1	0.75	0.482	0.662	3.53	de Haller < 0.72
Design point	0.5	Rotor 2	0.11	0	0.08	1	0.482	0.375	0.785	9.08	
First limit (peak)	0.42257	Rotor 1	13.88	0.344	0.2155	1.526	0.761	0.389	0.532	5.95	de Haller < 0.72
First limit (peak)	0.42257	Rotor 2	2.52	0.058	0.082	1.025	0.389	0.296	0.765	9.36	
Near choke (0.99 m_choke)	0.58663	Rotor 1	-3.21	-0.497	0.2873	2.035	0.756	0.834	1.09	3.47	
Near choke (0.99 m_choke)	0.58663	Rotor 2	0.04	-0.002	0.0801	1	0.834	0.57	0.707	9.07	de Haller < 0.72